**Politecnico di Milano**

**Master in Space Engineering**
**Academic Year 2022 / 23**

**Orbital mechanics course**

# Final assignment

# Interplanetary and planetary explorer missions


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**January 2023**

# Table of Contents

**Table of Contents**
**Assignment 1: Interplanetary Explorer**
**Assignment 2: Planetary Explorer**
- 2.1 Nominal orbit summary
  - 2.1.1 Assigned* parameters
  - 2.1.2 Assigned perturbations
  - 2.1.3 Other perturbations
- 2.2 Groundtrack
  - 2.2.1 Orbit GT
  - 2.2.2 Repeating GT
- 2.3 Orbit propagation
  - 2.3.1 Analytical vs. numerical integration
  - 2.3.2 Cartesian vs. Gauss's equations methods
    - 2.3.2.1 High frequency filtering
- 2.4 Evolution of the orbit
  - 2.4.1 Short term
  - 2.4.2 Long term
- 2.5 Real data comparison
  - 2.5.1 Object selection
  - 2.5.2 Parameter estimation
  - 2.5.3 Comparison vs. assigned model
  - 2.5.4 Comparison vs. our model

**Bibliography**

# Assignment 1: Interplanetary Explorer

> The PoliMi Space Agency is carrying out a feasibility study for a potential Interplanetary Explorer Mission visiting an asteroid in the Solar System, with an intermediate flyby on a planet.
>
> As part of the mission analysis team, you are requested to perform the preliminary mission analysis. You have to study the transfer options from Earth to the arrival asteroid, with a powered gravity assist (flyby) at the intermediate planet, and propose a solution based on the mission cost (measured through the total $\Delta v$).
>
> The mission departs from Earth, and the flyby planet and arrival asteroid have been decided by the science team. Constraints on earliest departure and latest arrival have also been set by the launch provider, the systems engineering team, and the Agency's leadership.

# Assignment 2: Planetary Explorer

> The PoliMi Space Agency wants to launch a Planetary Explorer Mission, to perform Earth observation.
>
> As part of the mission analysis team, you are requested to carry out the orbit analysis and ground track estimation. You have to study the effects of orbit perturbations, and compare different propagation methods. Also, you have to characterize the ground track, and propose an orbit modification to reach a repeating ground track (for better communications with our network of ground stations).

## 2.1 Nominal orbit summary

### 2.1.1 Assigned* parameters

| Semi-major axis | Eccentricity | Inclination | RAAN | Argument of periapsis | True anomaly at t=0 |
|---|---|---|---|---|---|
| 26,619 km | 0.7452 | 62.9089º | 60º | 30º | 0º |

**Initial date**: November 1st, 2021 at 0h 0' 0'' GMT

**Repeating groundtrack k:m ratio:** 2

***RAAN, argument of periapsis and initial true anomaly were left to our criteria, which were chosen in order to have a good movie and after fixing the initial date.

### 2.1.2 Assigned perturbations

- **$J_2$ effect** due to Earth's oblateness

$$\mathbf{a_{J_2}} = \frac{3}{2}\frac{J_2 \mu R_E^2}{r^5}\left[\left(5\frac{z^2}{r^2}-1\right)x\hat{\mathbf{i}} + \left(5\frac{z^2}{r^2}-1\right)y\hat{\mathbf{j}} + \left(5\frac{z^2}{r^2}-3\right)z\widehat{\mathbf{k}}\right]$$

$J_2$ = 0.0010826 as provided by the function *astroConstants*

- **Atmospheric drag** due to the absorption and reflection of massive particles against the spacecraft

$$\ddot{\mathbf{r}} = -\frac{1}{2}C_D\frac{A}{m}\rho(h,t)v_{rel}\mathbf{v_{rel}}$$
$$where: \quad \mathbf{v_{rel}} = \mathbf{v} - \omega_{\mathbf{E}} \wedge \mathbf{r}$$

The density at very high altitudes significantly depends on day/night cycles and the solar activity at any given time, which by itself makes the modelling of aerodynamic drag a challenge. There are several models, like the Jacchia-Bowman 2008, that adress this issue with varying levels of success. However, we are going to employ a simple exponential approximation that depends only on the altitude:

$$\rho = \rho_0 exp\left(-\frac{h-h_0}{H_0}\right) \; ; \; \rho_0 = 2\cdot 10^{-8} \; ; \; h_0 = 122km \; ; \; H_0 = H_0|(\rho = 3\cdot 10^{-12}, h = 400km$$

$C_D$ = 2.1

**A/m** = 0.0095 $m^2$/kg

### 2.1.3 Other perturbations

In the last point, the comparison with real data, we include an additional third model that features the following perturbations:

- **Atmospheric drag** (Explained earlier)
- **Solar radiation pressure**

$$\ddot{\mathbf{r}} = -\upsilon\phi_E C_R \frac{A}{m}\hat{\mathbf{r}}\ ;\ \phi_E = \phi_0\left(\frac{Rsun}{r_{sun}}\right)^2\ ;\ \phi_0 = \sigma T_{sun}^4$$

$$where:\quad \sigma = Stefan - Boltzmann\ constant, \phi = Solar\ flux$$

$$and:\quad \upsilon = shadowfunction(r, r_{sun}, R_E, R_{sun})$$

This shadow function is implemented and working and can be seen in the referenced book.

- **Earth gravitational model 1996** (And therefore substitutes the $J_2$-only perturbation)

$$U(r,\phi,\xi) = -\sum_{n=2}^{360}\frac{\mu R_E^n}{r^{n+1}}\sum_{m=0}^{n}(C_{nm}cos(m\phi) + S_{nm}sin(m\phi))\widetilde{P}_{nm}(\xi)$$

$$where:\quad r^2 = x^2 + y^2 + z^2\ ;\ \phi = tan^{-1}\left(\frac{y}{x}\right)\ ;\ \xi = \frac{z}{r}\ ;\ \widetilde{P} = norm.\ Legendre\,polynom.$$

Deriving -U(x,y,z) gives the perturbing accelerations we are seeking.

With this it's also possible to calculate the undulation potential, which shows Earth's uneven gravity field:

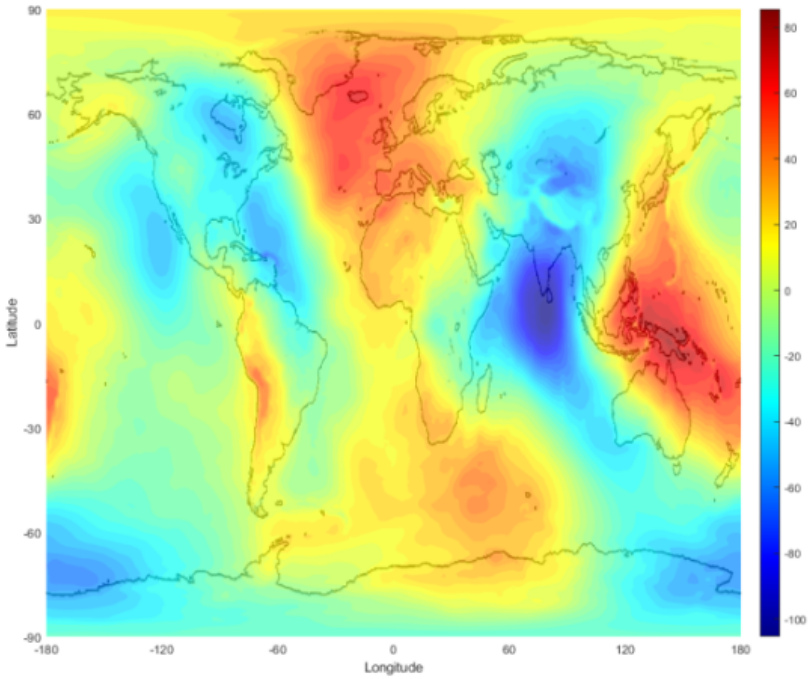


Fig.1 - Undulation potential T(lat,lon)

- **Sun and Moon 3rd body perturbations**

$$\ddot{\mathbf{r}} = GM_2\left(\frac{\mathbf{r_2} - \mathbf{r}}{\|r_2 - r\|^3} - \frac{\mathbf{r_2}}{r_2^3}\right)$$

Where **r** and $\mathbf{r_2}$ are the position vectors of the spacecraft and the planet of mass $\mathbf{M_2}$ relative to the Earth, **2** being the Sun or the Moon.

- **Relativistic effects**

$$\ddot{\mathbf{r}} = -\frac{GM_E}{r}\left[(4\frac{GM}{c^2 r} - \frac{v^2}{c^2})\mathbf{r} + \frac{4}{c^2}(\mathbf{r}\cdot\mathbf{v})\mathbf{v}\right]$$

## 2.2 Groundtrack

As a spacecraft orbits around a celestial body, the path drawn on it is deviated from an ellipse by the rotation of the body and the perturbations affecting the spacecraft.

### 2.2.1 Orbit GT

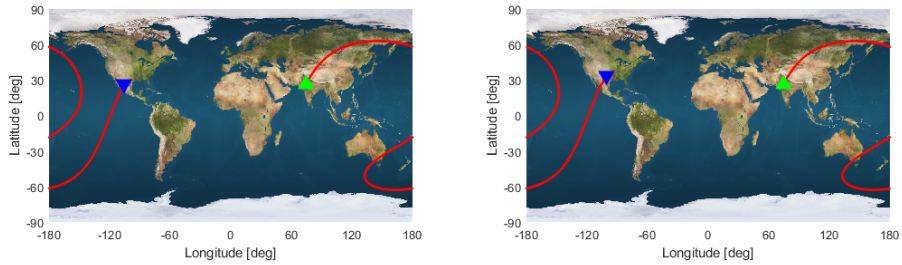


Fig.1 - 1 orbit groundtrack (perturbed-right)

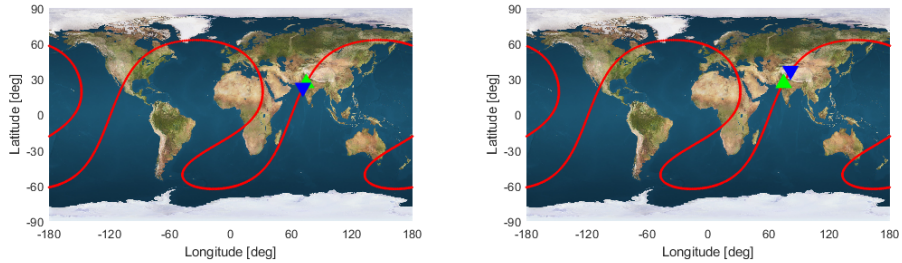


Fig.1 - 1 day groundtrack (perturbed-right)

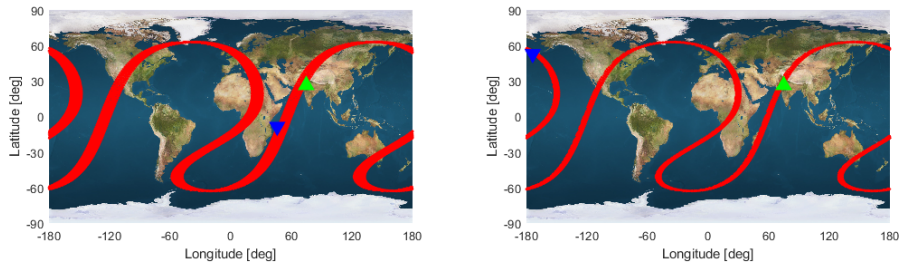


Fig.1 - 10 days groundtrack (perturbed-right)

### 2.2.2 Repeating GT

The orbit is already very close to having the k:m ratio of 2. Solve for the new semi-major axis from the equations:

$$T^2 = 4\pi^2 \frac{a^3}{\mu} \; ; \; \frac{T}{T_E} = \frac{m}{k} \; ; \; T_E = 23h\,56m\,4.1s$$

| Semi-major axis a | Repeating GT a | Delta |
|---|---|---|
| 26,619 km | 26,562 km | -57.236 km |

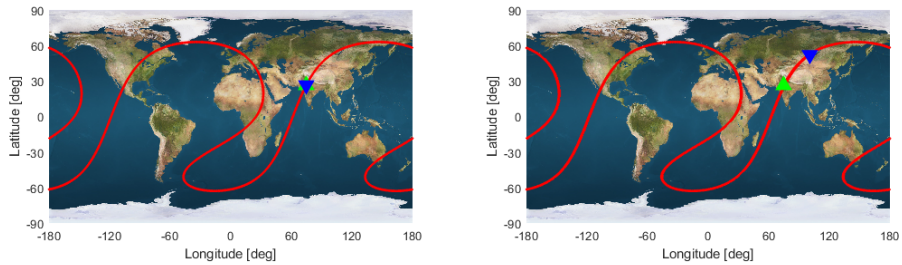


Fig.1 - k = 2, m = 1 (perturbed-right)

However, when propagated further into the future, the perturbed one deviates significantly:

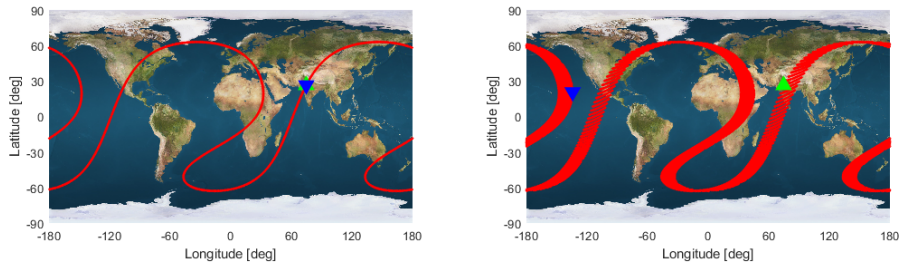

Fig.1 - k = 20, m = 10 (perturbed-right)

The aerodynamic drag reduces e but most importantly, a (smaller period, shorter gt), while J2 acts on the RAAN (shifted gt).

## 2.3 Orbit propagation

### 2.3.1 Analytical vs. numerical integration

### 2.3.2 Cartesian vs. Gauss's equations methods

#### 2.3.2.1 High frequency filtering

| Semi-major axis | Eccentricity | Inclination | RAAN | Argument of periapsis | True anomaly at t=0 |
|---|---|---|---|---|---|
| - | - | - | - | - | - |

The secular variations above have been obtained by applying a low-pass filter in the form of a moving window on the perturbed orbit. On one side, the J2 perturbation produces secular variations on the RAAN and argument of periapsis, and on the other, the drag acts as a negative impulse on the periapsis, gradually reducing both the eccentricity and semi-major axis.

## 2.4 Evolution of the orbit

### 2.4.1 Short term

We have made two videos of the first 3 orbits. First one shows the orbit from a fixed point relative to Earth's center:

https://www.youtube.com/watch?v=gYX6A3jnGZs

And the second shows the orbit from the point of view of the spacecraft, centered on Earth:

https://www.youtube.com/watch?v=YlP3BiDUItA

### 2.4.2 Long term

## 2.5 Real data comparison

### 2.5.1 Object selection

The aim was to find the satellite or debris with the closest 3 first orbital elements. After a search through some tracking websites, this is the result:

| | Semi-major axis | Eccentricity | Inclination |
|---|---|---|---|
| Assigned | 26,619 km | 0.7452 | 62.9089º |
| Real | 23,716 km | 0.712 | 63º |

The object is designated SL-6 R/B(2), a block-ML from a Molniya rocket that is orbiting the planet since 1999. It is catalogued in NORAD as 25850U and it's described as a 900kg, cylinder shaped (2.6x2.58x2.58m) rocket body, with an average cross section of 7.8824 and radar cross section of 2 sq. meters.

### 2.5.2 Parameter estimation

### 2.5.3 Comparison vs. assigned model

### 2.5.4 Comparison vs. our model

# Bibliography


Translational propagation LHS

MNIST dataset

Cost sensitive learning

Ensemble learning

GAN

Whittaker

Simulated Annealing

Particle Swarm Optimization

Genetic Algorithm

Saddle points in high dimensional data